Investigation of Lead-Acid Battery in the context of Automative SLI

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***Abstract*—Understanding lead acid batteries and their properties is empirical to progressing to better battery technologies and understanding how to develop batteries with varying properties for different situations. We tested various compositions and operating conditions for battery cells to investigate the effects on the voltage, power output and energy efficiency for the battery. Our results show that larger anodes, low currents/low discharge rate and high operating temperatures are good for energy efficiency, while large anode, high discharge rates and low temperature favour high power output. We also found that using a tin electrode gave better power output compared to our traditional lead acid cell.**

I. INTRODUCTION

SLI (starting, lighting, and ignition) batteries have been used in the automotive industry for years, as a result of their ability to deliver high bursts of power being perfectly suited to starting car motors. In comparison to deep discharge batteries they can provide provide much greater currents in smaller time frames, but suffer from great lifetime degradation when discharged deeply, and as a result are of often kept above 95% charge in real world applications. Understanding the central features to how these batteries operate is key to developing more efficient and longer life batteries with better charge and discharge properties, which is essential to the information age moving forward as more devices require batteries for different use cases. The lead-acid battery was first proposed by French physicist Gaston Plant in 1859[1]and is the most widely used SLI battery, consisting of a lead anode, some cathode, and an electrolyte solution composed of water and sulphuric acid. Over the years since it has seen small but steady improvements to its design.in 1881 Camille Alphonse Faure contributed the idea of using a lead grid lattice (anode) into which a lead oxide (cathode) paste was pressed to form a plate, creating an easier to produce battery[2]. In the 1930s gel electrolyte batteries started being used to reduce leakage and allow for a safer design, and then in the 1970s the valve regulated lead-acid battery was developed. We hope to make even more improvements moving forward. To do so, the properties of the battery that we need to consider are capacity, i-V characteristics, OCV, Energy Delivery and Efficiency through analysing charge and discharge curves and the respective capacities. We would also ideally analyse structural effects but that is outside of the scope of this article.

II. THERMODYNAMICS AND ELECTROCHEMISTRY

The Open Circuit Voltage (OCV) is a measurement of the cell voltage, and is derived from the two half cell equations that are found occurring at each of the electrode electrolyte interfaces. These equations describe the reactions at the anode and the cathode during the discharge of the battery, and their respective Voltages are measured in comparison to a Hydrogen-Platinum Standard Half Cell.

A. Oxidation Half equation - Anode

$$Pb + SO_4^{2-} \rightarrow PbSO_4 + 2e^- \tag{1}$$

$$E_1^0 = -0.351V$$

*Thanks to the rest of my 3.014 group, group 3!

B. Reduction Half equations - Cathode

$$PbO_2+4H^++2e^-+SO_4^{2-} \rightarrow PbSO_4+2H_2O \quad (2)$$

$$E_2^0 = 1.698V$$

The OCV (also sometimes referred to as the e.m.f) is the difference between these two values and is a measure of the potential difference the cell can generate.

$$E^0 = E_2^0 - E_1^0 = +2.049V \quad (3)$$

This theoretical value often is higher than the real experimental values as many factors will affect the cell voltage as we will see in the next section. The ionic compounds in these equations are transported through the electrolyte solution, and the electrons are carried via a wire between the anode and cathode. It is the electrons that do the electrical work when we connect the cell to an external device, so it is here where we measure the Voltage of the circuit. Without any load here it's called an open circuit, hence the name open circuit voltage.

Combined, these half cell equations give the overall reaction for discharging the cell, and is reversible through charging. Charging is done by applying a negative potential across the electrodes to force the movement of electrons back to their origin electrode.

$$Pb_{(s)}+PbO_{2(s)}+4H^+_{(aq)}+2SO_4^{2-} \rightleftharpoons 2PbSO_{4(s)}+2H_2O_{(l)} \quad (4)$$

The real world voltages of these cells are well described by the Nernst equation for the cell. Named after Walther Nernst, a German physical chemist who formulated the equation, the Nernst equation relates the activities of the species involved in the reaction, absolute temperature and molar quantities of the half cell with the standard half-cell potential to the half-cell potential in any condition. It is Derived from the Gibbs free energy of the reaction. as shown below. Gibbs Free energy can be used to calculate the maximum of reversible work that may be performed by a thermodynamic system [3].

C. Gibbs Free Energy to Nernst

Free Energy of Products - Free energy of reactants = Gibbs free energy of reaction

$$\Delta G - \Delta G^0 = RTln\frac{\prod_j a_{j_{products}}^{n_j}}{\prod_i a_{i_{reactants}}^{n_i}} \quad (5)$$

Where ΔG^0 is the Gibbs free energy in standard conditions, R is the gas constant, T is absolute temperature in K, and a dednotes the activities of the species. And given that $\Delta G = -nFE$ where F is the faraday constant, and n is the number of transferred electrons, for Open Circuit Voltage:

$$\Delta G - \Delta G^0 = -nF(E - E^0) \quad (6)$$

$$\therefore \Delta E = \Delta E^0 - \frac{RT}{nF}ln\frac{\prod_j a_{j_{products}}^{n_j}}{\prod_i a_{i_{reactants}}^{n_i}} \quad (7)$$

This is the equation known as the Nernst equation. Using this we can choose variables to change in our experiments that will affect the OCV, and think about how to achieve desirable properties for our batteries. We can vary E through cathode choice, measure variation with temperature, and also vary concentration of the electrolyte to see how activities affect our cell.

As well as Open Current Voltage, we ought to examine the energy efficiency of the lead acid cell through charging and discharging, measuring the capacity (amount of electrical charge a battery can store and deliver) and measuring the electrical work done by the battery through its energy output. in discharging. We can plot voltage against charge capacity and take the area under the graph of both the charge and discharge curves and use the ration to get the efficiency.

III. METHODOLOGY - I

We need to evaluate the desirable properties of Lead-Acid Batteries for our analysis.

A. Instruments

- HP 3457A Multimeter
- Topward DC PSU 3303D
- VWR Hotplate Stirrer
- Fluke 45 dual display multimeter

B. Battery Components

- Lead (Pb) and lead oxide (PbO2) electrodes from a Lichpower Model LTX9-BS 12volt 8-Ah Motorcycle Battery (Leoch Battery Co. Ltd., Shenzheng, China)
- Tin(Sn) and Pb-50wt%Sn alloy wires (Amerway)
- Pb (99.998%) foil 1.0 mm thick (Alfa Aesar)
- Sulfuric acid electrolyte (98%)(Mallinckrodt)
- 600 mL beaker with a PTFE (Teflon) cap to support the electrodes and a thermocouple
- lithium-ion battery (2.5 Ah, Milwaukee Tools, Redlithium 18650)

C. Experimental Layout

- OCV as a Function of Concentration
- Discharging the Battery at 3M with high and low current
- Charging the battery at 3M with high and low current
- Influence from Material and Size on Negative Electrode
- Temperature influence of cell voltage and energy efficiency
- Lithium-ion battery energy efficiency measurement
- energy efficiency measurement for different concentration sulfuric acid solutions

D. Battery Assembly and notes on operation

The PTFE cap should have two cylindrical holes to allow the threading of wire and the attachment of crocodile clips to hold the electrodes. There should be an additional hole for a thermocouple to pass through into the battery. A diagram will be place below to show the arrangement. Connect the negative black terminal of the multimeter to the Pb anode which can be identified by its dark grey colour, and connect the PbO2 cathode to the positive red terminal. This completes the lid of the beaker battery set up. The cap can then be added to the beaker containing to electrolyte to complete the battery. Each beaker was filled to approximately 230mL to ensure that the electrodes had good contact with the acid electrolyte. Ensure that the crocodile clips holding the electrodes are not in contact with the electrolyte, since this will affect the ion composition in the solution, and ensure that you have added the magnetic stirring bar before closing the battery. We will use a spin rate of 400RPM for the duration of the experiment to help with electrolyte transport and pushing the battery to its equilibrium point faster.

IV. METHODOLOGY - II

A. OCV as a function of concentration

Connect the cell to the multimeter ready to measure voltage. This experiment measures OCV as a function of electrolyte concentration.We will be using the following concentration of H_2SO_4, 0.01M, 0.1M, 1.0M, 3.0M, and 6.0M. These solutions were prepared just before the experiment for the sake of saving time, and are done by taking the higher concentration of acid (6.0M) and diluting it by adding acid to water in 5 separate 600mm beakers. 230mL of solution was prepared in each beaker. The cells were each connected, in turn, to the voltmeter and allowed to equilibriate with stirring on. It takes time for the electrodes to be soaked and become saturated with electrolyte maximizing surface area and producing consistent results. We waited for the voltage drift to be less than 0.5% of OCV per minute. The OCV was then recorded along with the temperature, and the cap of the cell was then removed, and the electrodes rinsed with deionized water before being placed in the higher concentration solution to create the next cell. We went from lowest to highest concentration to reduce any effects of residue acid within the electrode vastly affecting the concentration of the electrolyte it was being placed into. Ideally we should wash the electrodes in the molar acid concentration of the cell we are about to place them in but for this procedure we stuck with deionized water. It should be noted that the run-off for this experiment should be treated as hazardous waste, and collected and disposed of appropriately. Once all the concentrations had been tested, we then moved onto to testing dicharge rates and how the affect the batteries performance.

B. Discharging the Battery

To measure how rate of discharge affects the batteries performance, we configured the system using the 3M solution cell connected to both an ammeter and a voltmeter using the multimeters. The ammeter was placed in series with a variable

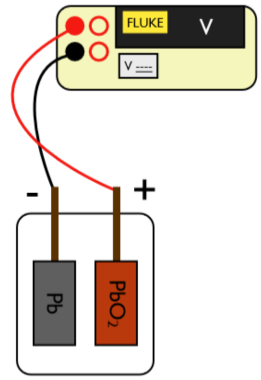

Fig. 1: OCV vs Concentration

resistor, and both the ammeter resistor path and voltmeter were connected in parallel as show below.

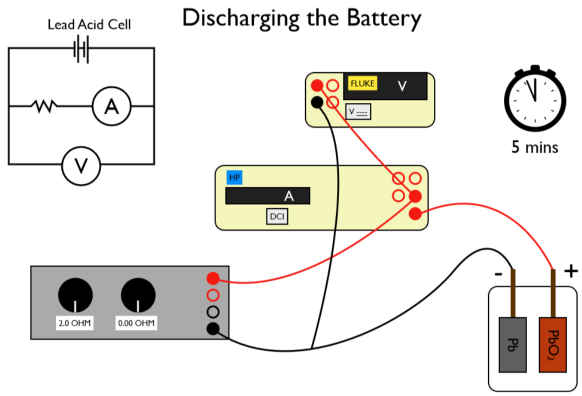

Fig. 2: OCV vs Concentration

The circuit was assembled and then left open at one of the points while the electrodes were allowed to equilibriate. The battery circuit was then closed and the variable resistor was then used to set the initial current in the circuits ammeter to a value of 0.2A, or 1.0A . Voltage and current were then recorded over a period of 5 minutes in 30 second intervals while leaving the resistance constant. The experiment was repeated for both current values. Temperature and the value of the resistor for each run was also recorded. We then moved on to measuring the properties of charging the battery with the same two currents.

C. Charging the Battery

The procedure for this experiment is much the same as the last. However the wiring for the measurements is different since we are now charging the Lead-Acid cell by applying some external votage. In place of the resistor in the previous experiment, we will connect the power supply.

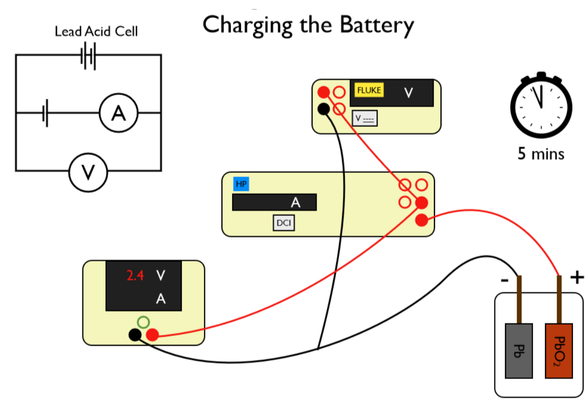

Fig. 3: OCV vs Concentration

The settings for which are a voltage limit at 3.0V, and the current limit set to either 1.0A or 0.2A as in the previous experiment. The circuit was then closed and values for voltage and current were then recorded over 5 minutes every 30 second interval. Temperature was also recorded. It is worth noting that we used the measurements from the multimeter to set out current to 1.0A and 0.2A respectively as the multimeter built into the power supply is not as close to the real world power delivered. Once the data for both current values had been recorded, we moved on to varying electrode material and size.

D. Influence from Material and Size on Negative Electrode

In this section we perform the experiment with 3M electrolyte at ambiet pressure, change the anode from the Pb plate from the battery to a smaller new peice of Pb (which should show the affect of surface area/electrode size) and will also analyse the same cell using a peice of Tin rod in place of the Pb plate. Our Pb piece is 4+-0.5cm in length (length is measured as amount of electrode submerged in the electrolyte, and was 0.125 inches deep. The diameter was 0.125inches from the manufacturers specification. Our lead piece was 6.8c+-0.1cm in length and was 0.4+-0.1cm wide. We measured the OCV for both electrode variants with the method in experiment A, and also performed the charge and discharge experiments with the same current values as in the previous experiments B and C.

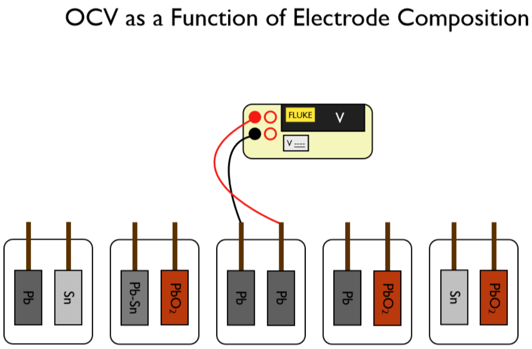

Fig. 4: OCV vs Concentration

E. Temperature influence of cell voltage and energy efficiency

Using the experimental set up identical to the one in experiment A, select the 3.0M electrolyte for the cell and allow to equilibriate, and then measure OCV as a function of temperature in 5°C intervals up to around 50°C (electrolyte temperatures measured by thermocouple). Note that due to hot plate stirrer will have to be set much higher than the cell temperature to allow the cell to reach the correct temperature in time.

Then using the same charge and discharge experimental setups as before, using a current value of around 1A, charge and discharge the Pb-PbS04 cell at 3m solution at an elevated temperature of around 55+- 5. Record the temperature and resistance value for the experiment and measure V and i every 30 seconds.

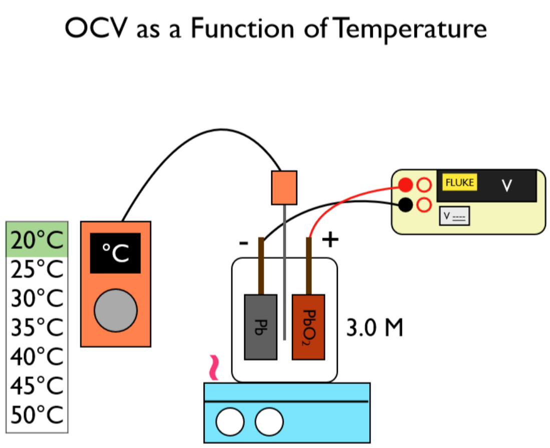

Fig. 5: OCV vs Concentration

F. Lithium-ion battery energy efficiency measurement

A lithium ion battery as listed in the equipment section has its voltage and current measured for charge and discharge at 1A every 30 seconds for a 5 minute period at room temperature. Record the temperature and resistance as well and include with the data.

G. energy efficiency measurement for different concentration sulfuric acid solutions

We also had data supplied to us by Ji for the Pb-PbO2 1.0M electrolyte charge and discharge at 1.0A and 0.2A ,as well as the same cell with a 6.0M electrolye.

V. Results and Analysis

A. Analysing affects on OCV

Relate data for lead acid with nernst As we saw in our thermodynamics section, the batteries properties can be described in part using the nernst equation, which gives us our theoretical value for OCV. From the equation, we can see that OCV should scale linearly with temperature, the natural log of the ratio of the activities, and inversely with the number of moles.

Plot OCV vs Molar Conc for sulf at ambient

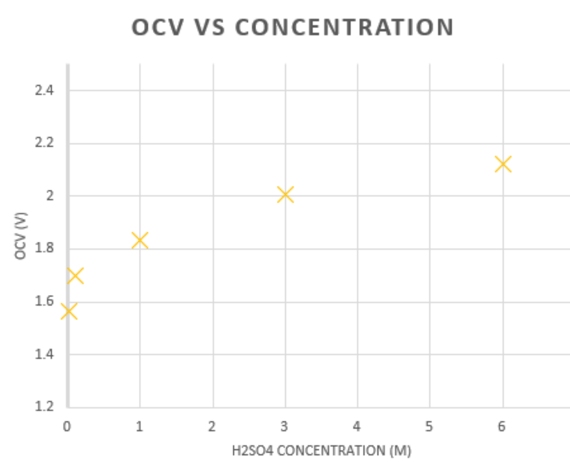

Fig. 6: OCV vs Concentration

Our graph for OCV vs molar concentration (Fig 1.) shows a logarithmic looking trend. This agrees with our Nernst equation, since concentration of electrolyte should be proportional to the activities.Knowing this and seeing this from the data, we know that when it comes to choosing the electrolyte concentration for the battery that the increase in OCV from increasing molar concentration quickly falls off, so choosing a concentration that is both operationally viable and cost effective is feasible. Choosing significantly higher molar concentration of H_2SO_4 would give a neglibile increase in OCV and would thus not be worth the trade off in safety and ease of manufacturing.

Plot OCW vs ln(activities) at ambient, fit points , estimate E0

Plot OCV vs temp, what data from slope?

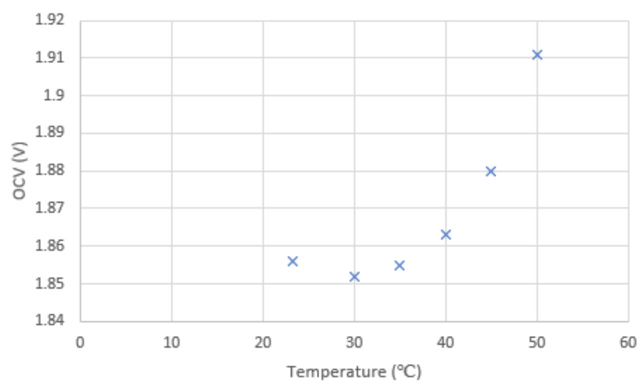

Fig. 7: OCV vs Temperature

Fig 2. is the plot from out data in experiment E for OCV vs Temperature. As we can see there is an overall increase in OCV as temperature increases. However these plots would seem to fit a non linear curve which the gradient of is increasing over time. This disagrees with our theory since we should expect OCV to vary Linearly with temperature from the theoretical value given by the half cell equations. One possible explanation for this is that before measurements started being taken, the cell had not been given enough time to reach an equilibrium state. This could be due to the electrodes not yet being fully saturated with electrolyte for example, having been rinsed off after the previous experiment. This would explain the increasing gradient since not only is the OCV increasing from the increase in temperature providing more energy, but the electrodes themselves are able to operate more effectively over time. Further work could be done here to analyse this deviation from theory.That aside, the trend does show that with increasing temperature, OCV does increase.

B. Power Properties

Our most important thing to consider for our real world use cases is power that the battery can supply, and how the variations in cell composition and operating temperature affect this property.

1) Power vs Temperature: Plotting our data for 3M electrolyte at two different temperatures [fig 8.], we can see that on average the cell at ambient temperature delivers approximately 0.9W more power. There is an initial large increase in the curve which is likely due to the electrodes still equilibriating/wetting in the cell. Once this stabilizes we can see that both cells decrease in power over time at approximately the same rate. As the cell discharges it is able to deliver less power, but the relative amounts of power differs with operating temperature.

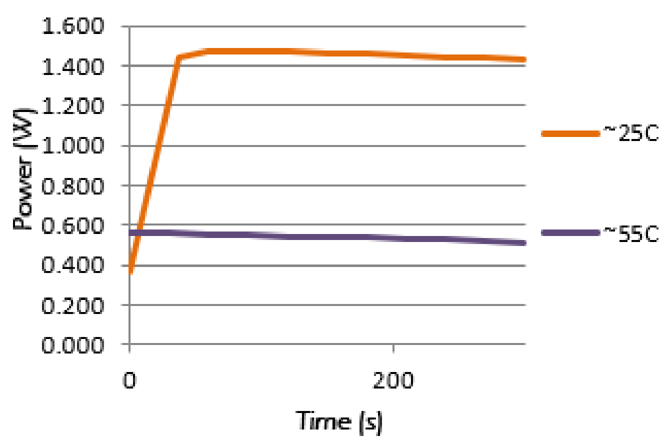

Fig. 8: Power with varied Temerature

2) Power vs Composition: In fig. 9 We have plotted our graphs for power vs composition. The compositions of the cells graphed are Pb(large electrode), Pb(small electrode), Sn and Lithium ion. This data shows that the power delivered by the

Sn electrode in place of the Pb (small) electrode is much higher for a similarly sized electrode. This result is significant and should be noted. The large Pb electrode gives a higher power rating than the small Pb electrode as predicted from our theory due to the increased activity from the increased amount of electrode exposed to electrolyte. The Lithium ion battery in comparison to all the other cells gives a significantly higher power output but does show more decrease over time as it discharges.

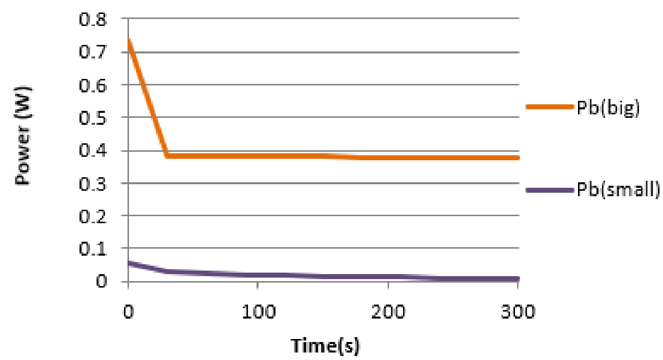

Fig. 10: Power for Differing electrode size

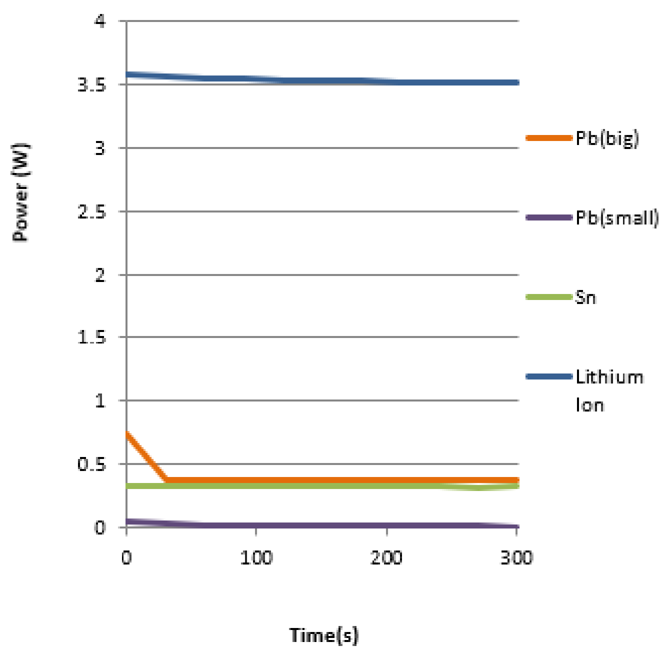

Fig. 9: Power with varied electrode composition

3) Power vs Electrode size: Following from our previous point, we have also plotted just the small and large lead electrodes against each other [Fig. 10] to see at a closer level the difference between the two. The trends are the same, the larger electrode giving a significantly higher power output. It should be noted that there is a sharp decrease in power output with the large electrode that seems outside of the norm. This could be an anomaly in our data but more investigation should be done here.

4) Power vs Discharge rate: By varying the resistance in out circuits, we were able to control the current from the battery and thus the discharge rate. By plotting the power over time for different current resistance values we can see the effect on discharge rate on power [Fig. 11]. For our 1.3Ohm plot, there is a steep increase given by the initial point being 0.4W. This should be ignored as it is likely due again to intial fluctuations and electrode wetting/equilibriating. The cell with the lower resistance (1.3 Ohms) and thus higher discharge rate dropped its power output faster than the cell with the higher resistance (9.0 Ohms).

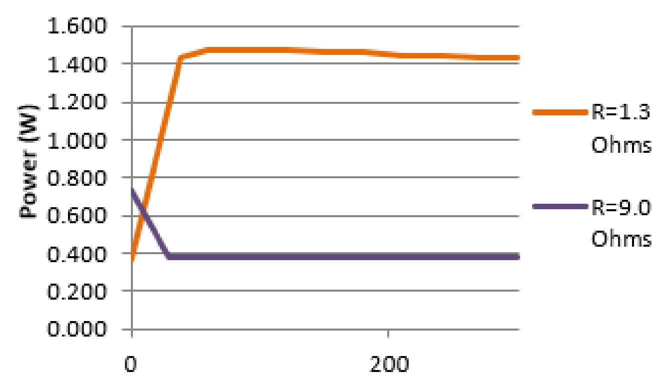

Fig. 11: Power with discharge at different resistance

C. Energy Efficiency

Energy efficiency is essential to understand for our cells. How much return of our initial energy that we put in is important to minimize waste and know realistically how much usage we are going to get out of the cells. Using the formula for efficiency outlined in the thermodynamics section we calculated the efficiencies for the different cell compositions and tabulated them below.

Electrode Efficiency's			
Electrodes	Current (A)	Temp (°C)	Efficiency (%)
Pb(big) & PbO2	1.09	24.4	52.1
Pb(big) & PbO2	0.19	24.4	64.2
Pb(big) & PbO2	0.18	54.8-58.8	135.0
Pb(small) & PbO2	0.04	24.4	13.7
Sn & PbO2	0.17	24.4	66.4
Lithium Ion	1.02	24.4	91.7

VI. Conclusions

Our results show that larger anodes, low currents/low discharge rate and high operating temperatures are good for energy efficiency, while large anode, high discharge rates and low temperature favour high power output. We also found that using a tin electrode gave better power output compared to our traditional lead acid cell. This lines up with what we expected from our thermodynamic analysis. In the context of SLI systems, we would expect to see the high power output configuration being favoured since the application requires a high power output in a short period of time but with low discharge percentages from the full storage capacity, meaning the undesired properties that come with the high power configuration do not negatively impact the application. The use of tin electrodes also favours these applications by providing high power with a smaller electrode than the lead, but due to the mechanical properties of pure tin and its tendency to phase change around room temperatures and below it would not be suitable for automotive use where in cold weather the battery may experience temperatures that drastically would affect the electrode.

VII. Further Work

Further work should also be done to investigate the effects of cycling the batteries on their capacities, and other alloys as potential electrode candidates. These preliminary results we have obtained should also be carried out over a longer time frame to ensure the cells have reached equilibrium after being put together before testing to ensure reliable results, since in ours we can see the effects of the non equilibrium state drastically affecting OCV and power output.

ACKNOWLEDGMENTS

Thanks to Dr Ji Zhao, Prof. Donald R. Sadoway, Prof Kimerling, Dr. Quanquan Pang, Mr. Jiashen Meng the rest of Group 3 for making this study possible.

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